SPE-229113-MS

Attesting Clean Energy at the Source: A Cryptographic Framework for Real-Time Renewable Tokenization

Hans Rempel, Diode, Sheridan, WY, USA; Mark Smith, CleanConnect, Windsor, CO, USA

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Clean energy (energy from sources that emit negligible pollution or greenhouse gases during production) reduce harmful environmental impact, making it essential achieving sustainable development goals. Although natural gas is not normally considered clean energy, carbon intensity (CI) scores can vary from 10 to over 80 gCO2e/M across producers. As Empirical Natural Gas (ENG) emerges as a leading methodology to capture ISCC certified feedstock attributes, including Energy/Environmental Attributions Certifications (EACs), they have become a clean energy commodity tradable independent of the physical constraints of the underlying natural gas. As producers, buyers, and traders embrace ENG as an efficient and cost competitive tool to achieving feedstock CI targets independent of regional location, the audit initiation and renewal process is time consuming, lacks repeatability, and ultimately is not real-time enough to fully underwrite the risk of double counting or other potential anomalies in EACs. This latency and opacity hinders the industry's ability to track, trade, and optimize energy assets efficiently in real time.

By drawing from proven blockchain incentive structures, this study introduces a novel security framework to provide protections for the various parties involved in carbon certification from production to use. The proposed architecture represents one of the first practical implementations linking edge-level cryptographic attestations with decentralized clean energy markets, providing a trusted, automated foundation for scalable sustainability metrics and carbon reduction.

Introduction

Energy buyers face mounting challenges in meeting sustainability goals while ensuring reliable and affordable supply. The emergence of Energy Attribute Certificates (EACs) is decoupling regional producer/supplier constraints by creating a tradable commodity that can be used to achieve a target carbon intensity (CI) score. This is transformative, particularly in the context of natural gas, which has sometimes been dismissed as a "transitional" or even "dirty" fuel. Through careful quantification of CI at the producer level, EACs recast natural gas as a differentiated, premium product whose quality can be defined on a granular basis independent of region. This is revolutionizing energy trading.

The fundamental value proposition of EACs is twofold:

- **For Producers**: EACs open up access for premium product to buyers outside their immediate geography, often in jurisdictions with stringent decarbonization policies (e.g., EU, Japan), who are willing to pay significant premiums for low-CI gas. This redefines the business model for upstream operators, shifting them from unsophisticated price takers to strategic, differentiated suppliers.
- **For Buyers:** EACs offer the power to precisely meet or undercut regulatory CI thresholds regardless of the regional mix or the availability of physical infrastructure, thereby decoupling environmental performance from physical logistics. This supports compliance with policies like the EU's Carbon Border Adjustment Mechanism and emerging clean fuel standards across Asia.

However, as trading volumes and velocity scale, so does the risk of "double-claiming," delays in verification, and erosion of trust. Effective scaling is only possible if the speed and integrity of audit and certification systems keep pace, transitioning from slow, retrospective processes to automated, real-time digital controls. The transition to scalable, auditable EAC systems underpins the credibility, liquidity, and growth potential for global clean energy commodity markets.

The clean energy commodity market can look to the concepts of "real world assets" (RWAs) and "actively validated service" (AVS) frameworks pioneered in the blockchain and Web3 infrastructure industries for inspiration on how to introduce dynamically enforced reliability and integrity of digital commodities at scale. In AVS systems, core market actors—such as node operators, validators, or service providers—stake financial collateral and are continuously monitored by automated protocols that validate their real-time performance. If a service anomaly, outage, or breach is detected, automated, and sometimes manual, mechanisms can penalize (slash) the staked value of the underperforming party, while also rewarding those who help maintain system integrity. This approach blends real-time auditability with automatic, high-frequency settlement and incentivization, thereby deterring malfeasance and aligning all actors to robust, verifiable outcomes.

Translating this methodology to clean energy commodities, EAC registries and carbon markets can embed similar "smart contract" based programmable incentive and penalty systems in certification frameworks. Producers, auditors, and traders would not only undergo conventional checks but would also be required to lock value in escrow against their performance and conformance to protocol requirements. A combination of automated and manual surveillance would check attribute accuracy, chain-of-custody integrity, and timely certification updates. In the event of double-claiming, data tampering, or delayed reconciliation, the responsible parties would face locking of the underwritten amounts and potential loss of their some or all staked value (the slashed amount). The slashed amount can then be redistributed to affected buyers, honest participants, or to a market integrity fund. By leveraging these programmable incentives—already field-proven in blockchain infrastructure—clean energy commodity systems can foster an environment of continuous vigilance and transparency, sharply reducing trust gaps and scaling market integrity with unprecedented efficiency.

Empirical Natural Gas as an Evolution of Carbon Insets

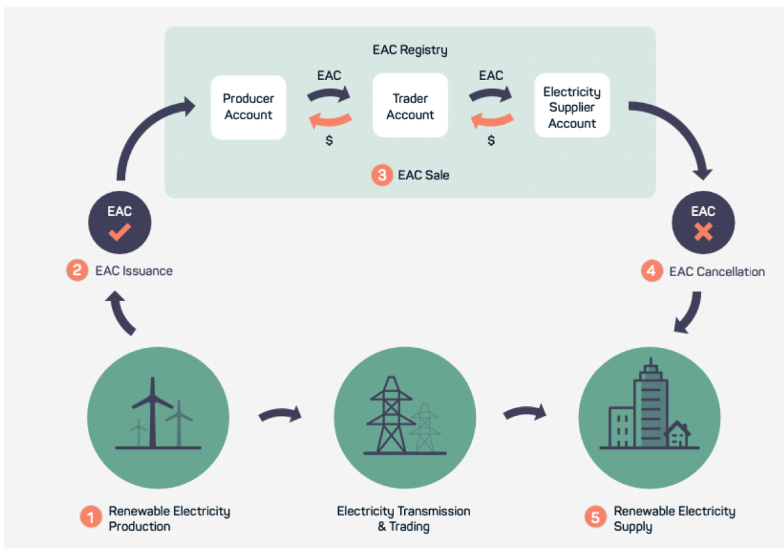
Traditional decarbonization strategies in gas supply chains have relied on carbon offsets—after-the-fact credits representing emissions reductions or removals, typically sourced from different sectors or even continents. These offsets are prone to criticism for lack of immediacy, traceability, and alignment with the operational realities of a gas buyer's own supply chain.

By contrast, carbon "insets" substitute lower-carbon-intensity (CI) fuel streams into the supply chain itself, achieving emissions reductions at the source. For natural gas, this means that instead of purchasing generic, high-CI pipeline gas, buyers contract for empirically validated, low-CI molecules. These molecules are traced through the supply chain using Energy Attribute Certificates (EACs) and advanced data infrastructure.

Real-Time Source Attestation of EAC Attributes

Empirical Natural Gas (ENG) harnesses real-time data collection devices - such as IoT sensors, continuous emissions monitoring systems (CEMS), and advanced supervisory control (SCADA) - to document CI at the site and batch level. This granular measurement system, when coupled with mass-balance certified EAC issuance and blockchain registries, ensures that each unit of ENG is accompanied by a digitally verifiable, time-stamped EAC. These EACs track not just the source and physical attributes but also the exact window of production, making it possible to transact and retire attributes in near real time. Such hourly or sub-hourly attestation is vastly superior to traditional annualized REC models, reducing the risk of "double-claiming" and increasing market confidence.

Hourly Granular EAC Framework and ENG Source Attestation.



The diagram illustrates IoT-based real-time CI measurement at a gas production site, the creation of uniquely timestamped EACs, their transfer over a blockchain registry, and final retirement by an end buyer.

Notably, real-time ENG-based EACs help buyers achieve transparent hourly Scope 2 market-based reporting, as now called for by GHG Protocol best practices and supported by leading industry members (e.g., Google, Microsoft). This enables both regulatory and voluntary claims with an unprecedented level of auditability.

Economics of ENG: Cost-Effectiveness and Market Impact

The economics of ENG are equally compelling:

- **Cost Efficiency**: Empirical certification at the source enables a significantly expanded pool of "clean" gas compared to RNG alone, bringing liquidity and price discovery to the market. RNG is supply-constrained, infrastructure-intensive, and can exceed \$15–20 per MMBTU (up to \$200 per tCO2e abated). By comparison, ENG blending with certified low-CI natural gas often achieves comparable or lower CI at approximately \$7–14 per MMBTU, reducing overall abatement costs to the \$70–100/tCO2e range—close to those seen for solar and wind.
- **Blended Product Optimization**: With real-time CI scoring, producers and traders can blend ENG and RNG to meet exact compliance targets for buyers—such as those needed by refineries, AI datacenters, or global shipping—often at half the cost of RNG-only strategies. This blending supports global climate goals while retaining the reliability and infrastructure compatibility of natural gas.

- **Incentive Alignment**: Automated, auditable frameworks for ENG attestation provide strong confidence for voluntary and compliance carbon markets. Independent, empirically verified CI scores—recorded at source and substantiated in real time—virtually eliminate the misallocation, mispricing, or double-counting that have hampered past carbon credit regimes.

Levelized Costs (2024 $/MMBTU) and CI Scores.

Compiled from FortisBC, Wood Mackenzie, Nature Communications

Product	Levelized Cost ($/MMBTU)	CI (gCO2e/MJ)	Notes
Pipeline Gas	6–8	75–85	High emissions factor, low cost
Landfill RNG	15–20	40–45	High cost, moderate CI reduction
ENG (Empirical)	7–14	10–20	Low CI, cost competitive, highly traceable
Solar/Wind	N/A	0	Not compatible with gas infrastructure

In summary, Empirical Natural Gas sets a new standard: it provides real-time, source-level attestation of clean energy attributes and cost-effective decarbonization for gas buyers. By integrating credible, empirical measurement systems with granular, blockchain-based certification tools, ENG paves the way for robust, transparent, and economically advantageous carbon insets—delivering on both climate performance and bottom-line results.

Participants and Market Roles: Producers, Traders, and Buyers in a Transformed Ecosystem

The empirical EAC ecosystem integrates three core market actors - Producers, Traders, and Buyers - each independently optimizing for their goals, yet capable of forming mutually beneficial relationships that underpin a robust and dynamic clean energy market.

Producers: Pioneers of Empirical Certification

Producers lead by employing advanced empirical measurement, monitoring, and reporting technologies (e.g., IoT-based sensors, continuous emissions monitoring, and digital twins) to quantify carbon intensity at the source. By securing third-party certifications such as ISO 14067 or ISCC, producers unlock premium pricing and expand access to international clean fuel buyers. This analytical rigor transforms emissions control from a regulatory burden into a source of commercial differentiation and value creation. Importantly, the credibility and granularity of their data serve as the foundation upon which the rest of the market operates —helping traders and buyers trust the attributes being traded.

Traders: Market Makers and Relationship Builders

Traders - including specialist commodity firms - bridge the gap between disparate producers and buyers by aggregating, blending, and optimizing gas streams and EACs to match a wide variety of buyer specifications. They provide liquidity, standardize complex contracts, and manage compliance across markets. Traders are also well-positioned to underwrite risk: by verifying the provenance and certification of each EAC (often using blockchain-backed registries and digital contracts), they ensure transaction integrity and ease counterparty fears around double-counting or non-compliance. By creating flexible product offerings and optimizing logistics, traders add value for both upstream suppliers and downstream customers, functioning as integrators and trust anchors for the marketplace.

Buyers: The Decarbonization-Driven Demand Side

Buyers represent energy-intensive industries (steel, chemicals, shipping, data centers, and utilities), many of whom are motivated not just by compliance, but also voluntary sustainability commitments and investor expectations. With EACs, buyers can precisely meet or exceed regulatory Scope 1, 2, and 3 requirements, often beyond the physical limitations of local energy infrastructure. The ability to source and then retire EACs from a global pool—rather than relying on local supply—gives them agility and strategic leverage in meeting increasingly stringent decarbonization targets.

Mutually Beneficial Relationships: Collaboration and Shared Value

Co-Creation and Premium Differentiation. Producers and traders can work in partnership to co-design data collection frameworks, blending strategies, and verification protocols. By aligning on high-fidelity measurement and transparent chain-of-custody systems, they jointly unlock access to high-value buyers.

The resulting data transparency reduces certification costs, accelerates audits, and cultivates buyer trust, enabling all actors to secure a larger share of clean energy premiums.

Risk and Value Sharing. Innovative contract structures—such as split-the-savings agreements or risk-sharing pricing—allow market participants to share both upside and downside in price volatility or regulatory changes. For example, traders and buyers may co-invest in new measurement systems at the production site, sharing both the initial cost and future gains derived from higher-certainty or lower-carbon attributes. Platform-based trading models allow for transparent, continuous market interactions, further aligning stakeholder incentives.

Joint Market Development and Policy Advocacy. With each role optimizing individually yet interacting frequently, the ecosystem benefits from collaborative advocacy for market standards, regulatory harmonization, and technological best practices. Producers, traders, and buyers often join industry working groups, supporting the development of new EAC standards, digital registries, or green supply chain frameworks, which not only ensure market credibility but unlock new global demand.

Transparency, Trust, and Scaling Impact. The integration of empirical measurement, blockchain-based registry systems, and rigorous third-party audits links technical rigor with market credibility. By pooling expertise - producers on data, traders on contracts, buyers on standards - mutual accountability and trust are continuously reinforced, reducing barriers to large-scale uptake and cross-border trading.

In summary, the empirical EAC market is a sophisticated ecosystem where producers, traders, and buyers —while each optimizing for their own priorities—create positive feedback loops of value, credibility, and innovation through collaborative relationships, risk sharing, and real-time, data-driven transparency. This synergy propels the scaling and integrity of global clean energy trading far beyond what any individual stakeholder could achieve alone.

Auditing Challenges: The Limits of Conventional Certification and Market Risk

The credibility and functioning of Energy Attribute Certificates (EACs) fundamentally rely on the rigor, reliability, and frequency of the audit and certification process. Leading programs, such as the International Sustainability and Carbon Certification (ISCC), provide comprehensive frameworks for mass-balance tracking, chain-of-custody, and third-party audits that span the journey from wellhead to end-consumer. Each participant—producers, traders, and buyers—must maintain detailed, auditable records consistent with international standards like ISO 14067. Despite these robust frameworks, ISCC and similar certifications remain rooted in a periodic, physical audit paradigm—an approach that often cannot keep pace with the speed and scale of modern, digitally traded EAC markets.

Market Exposure and Systemic Risk

A key vulnerability emerges from the temporal lag between audits: EACs may be transacted hourly or continuously in integrated digital markets, but audit verification often occurs only once per year. This creates a "liability window" in which large volumes of EACs—potentially valued in the tens or hundreds of millions of dollars per month—are traded on the strength of certifications that could have degraded or been misrepresented. If an audit fails, or if discrepancies later come to light, downstream parties are collectively exposed to substantial financial, contractual, and reputational risk.

A recent academic investigation into the voluntary carbon markets quantifies these risks: among nearly 100 projects initially certified by major registries, subsequent rejections or identified over-crediting cast doubt on over $1 billion in credits, directly undermining buyer trust and project valuation. One high-profile case involved a forestry offset scheme where a failed audit led to the invalidation of over $180 million in issued credits—a financial shock that also triggered cascading legal and reputational fallout for project developers and buyers. The research shows that project developers often select and pay their own auditors, introducing structural conflicts of interest.

This results in systemic over-crediting and "rubber-stamping" of non-additional or impermanent claims, as evidenced by the large number of subsequent suspensions and rejections from major registries.

Trader as Key Fiduciary in Automated Blockchain-Based Auditing

As commodity and EAC markets shift toward rapid digital trading, the role of the trader evolves dramatically. Traders—no longer mere intermediaries—become fiduciaries and trust anchors in a new regime of automated, blockchain-enabled, Actively Validated Service (AVS) reputation and audit systems. In distributed ledger or AVS frameworks, traders are responsible for:

- Aggregating and blending certified attributes from multiple producers, ensuring that all records are traceable, real-time, and fully reconciled on-chain.
- Verifying attribute provenance to trigger real-time compliance checks and automated reconciliation against auditor-issued certificates.
- Acting as fiduciaries for buyers and producers by underwriting the integrity of transactions and managing producer relationships when a slashing mechanism is triggered.

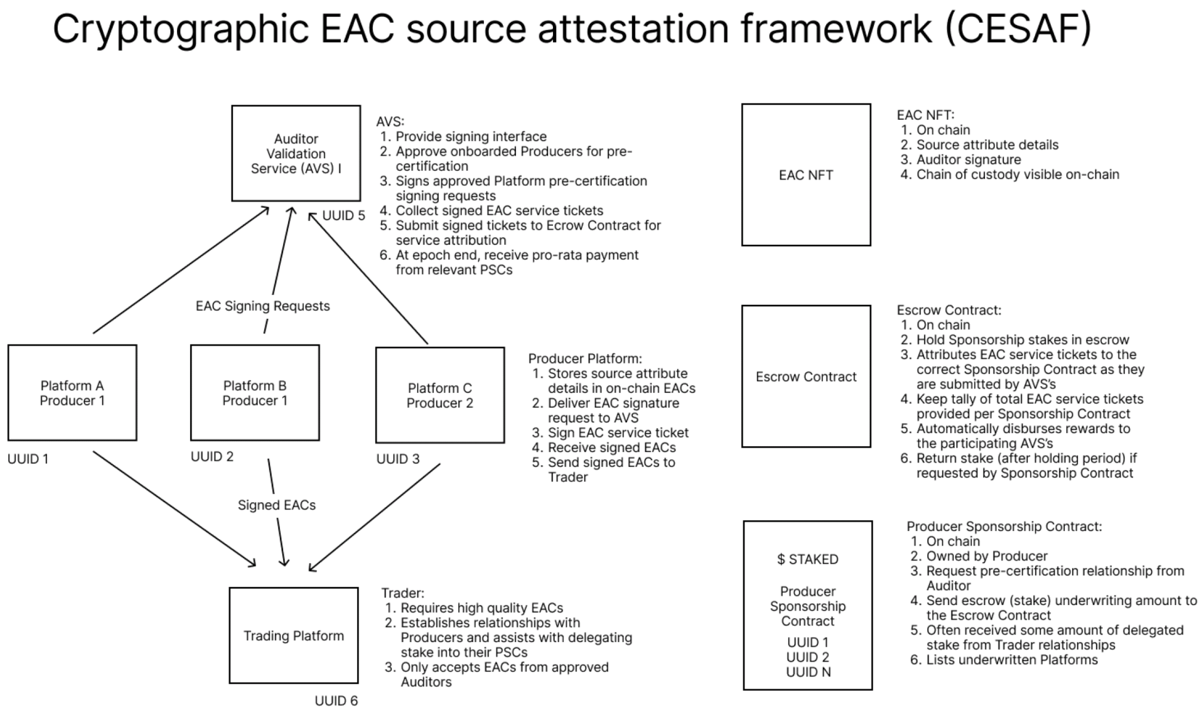
The trader's fiduciary responsibility is not merely operational; it is increasingly codified into smart contracts and risk protocols, ensuring that any failure—audit or otherwise—triggers corrective actions and minimizes the propagation of invalid claims through the marketplace. In the AVS model, producer source attribution, auditor approval, lot-level traceability, and chain of custody are all immutably recorded and financially underwritten, fostering trust and unlocking broader access to institutional investment and participation.

Real-world blockchain deployments in the energy sector, such as the Iberdrola blockchain pilot project, have demonstrated that integrating auditability and traceability at key stages of the supply chain can radically decrease both financial risk and market latency. Automated systems with instant reconciliation, immutable records, and trigger-based alerts address the weaknesses found in legacy annual audit cycles, substantially reducing the risk and magnitude of market failure.

Cryptographic EAC Source Attestation Framework (CESAF): Market-Based Trust for Clean Energy Markets

Digitalization is fundamentally transforming the security, efficiency, and credibility of Energy Attribute Certificate (EAC) markets, laying the groundwork for global trust and high-integrity climate impact. The Cryptographic EAC Source Attestation Framework (CESAF), as depicted in the following diagram, is a paradigm-shifting architecture that leverages blockchain technology, smart contracts, and automated cryptographic validation to address the weaknesses of legacy, paper and annual audit-based systems.

CESAF Block Diagram.



The image captures a block diagram of the CESAF system as inspired by blockchain Actively Validated Service underwriting for the clean energy commodity market. Each core market actor, as well as the Auditor, is represented. Refer to the sections below for more information.

On-Chain Attribute Certificates (EAC NFTs)

At the heart of CESAF, each unit of clean gas or verified low-carbon energy is assigned a unique digital certificate represented as a non- fungible token (NFT) on a blockchain ledger. These EAC NFTs embed all relevant environmental data, provenance, ownership transfer, and compliance records, creating an immutable, tamper-resistant digital chain of custody. The use of blockchain for attribute tracking assures market participants that certificates are auditable on demand and resistant to double-counting or unauthorized manipulation.

Third-Party Pre-Certification via Auditor Validation Service (AVS)

A crucial innovation in CESAF is the introduction of automated, auditor-operated Validation Services (AVS). These AVS systems review and digitally sign EAC NFTs before market entry, confirming that:

- The NFT was issued by an accredited, onboarded EAC source attribution platform
- The associated Producer Sponsorship Contract (PSC) is valid and collateralized
- Escrow funds back the EAC, covering potential compliance breaches

This AVS pre-certification step provides cryptographic assurance to traders and buyers that each NFT is genuine, standards-compliant, and financially underwritten - enabling real-time market due diligence and reducing reliance on infrequent manual audits.

Escrow Mechanisms, Slashing, and Incentive Alignment
Every EAC NFT in this framework is underwritten by value locked in a smart-contract escrow, as visualized in the CESAF diagram. If subsequent audits or automated surveillance identify anomalies, double-claiming, or misreported attributes, the escrowed value can be "slashed" (partially or entirely seized) and redistributed to injured parties or the governing market authority according to pre-established rules. This mechanism directly aligns the incentives of producers, traders, and auditors with market integrity: parties stand to lose financially if they allow errors or fraud, but benefit from maintaining robust, verifiable standards.

Payment and Workflow Automation
With every escrow period end, payment to auditors is triggered automatically when their AVS signatures remain uncontested, ensuring rapid turnover and revenue flows that scale transparently with transaction throughput. This replaces manually mediated, invoice-driven compensation with trustless, programmable settlement, thereby improving the speed and reliability of market oversight.

Radical Transparency: Immutable, Auditable Markets
By storing all relevant events - EAC minting, AVS signatures, escrow/PSC status, transfers, and retirements - on a ledger, CESAF ensures radical market transparency. Any regulator, market participant, or researcher can independently verify certificate provenance, compliance history, and transaction context in real time, drastically reducing opportunities for backdating, double-claiming, or manipulation. This transparency is crucial for institutional buyers and for linking EACs credibly to emerging climate disclosure and ESG frameworks.

Scalability and Market Impact
Because CESAF automates validation, settlement, and penalty processes, it enables real-time, high-volume, cross-border trading of EACs, with risk management and market integrity enforced "on-chain." Empirical studies show such blockchain-based certification and attestation frameworks dramatically reduce market frictions, audit latency, and the frequency and magnitude of compliance failures. The result is a resilient, adaptive infrastructure ready to support the multi-billion-dollar clean energy markets of the coming decade.

Conclusion
The Clean Energy commodity market must adopt source EAC attestation and use market incentives to streamline the audit process. Empirical EACs and digital attestation frameworks like CESAF are not simply technical upgrades—they are foundational to the next era of global sustainability. They repackage environmental performance as a tradable, verifiable, and bankable commodity, unlocking new financial flows and fundamentally reshaping incentives throughout the value chain.

- **For Producers:** New revenue streams, higher market values, and a competitive advantage for operational excellence.
- **For Traders:** Deeper liquidity, risk management, and product customization in clean energy supply.
- **For Buyers:** Reliable, auditable decarbonization, unchained from the limitations of local energy mixes or grid attributes.

As energy systems digitize and democratize, trust in the underlying real time data becomes both a commercial and a regulatory imperative. By embracing a system for real time source attribution and financial underwriting for auditor-approved EACs, the industry can optimize for clean energy across the global ecosystem.